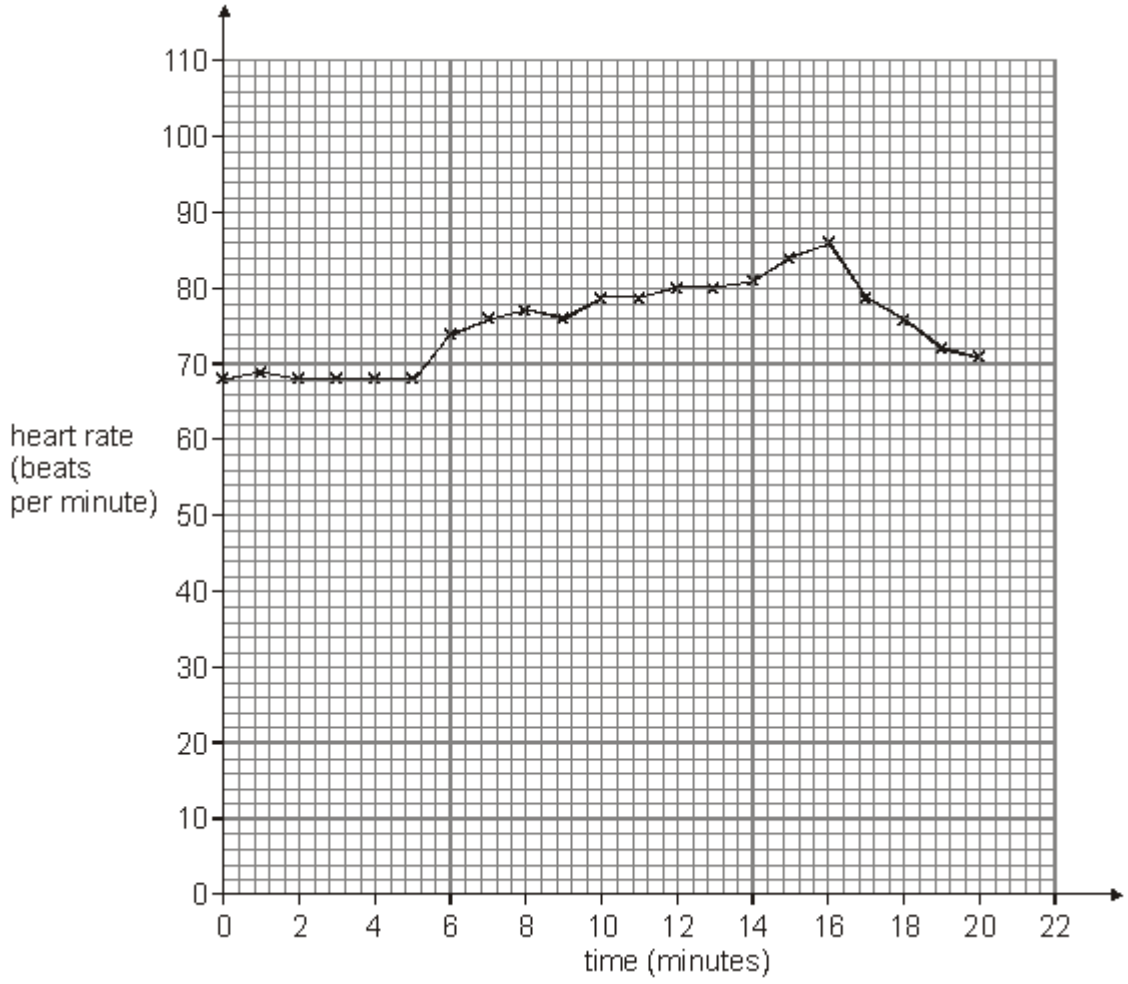
Q1.

Harry investigated the effects of fizzy cola drink on his heart rate.

First he measured his heart rate every minute for 5 minutes when sitting down. Then he drank some cola.

He continued to measure his heart rate at regular intervals.

This is a graph of his results.



(a) Why did Harry measure his heart rate every minute for 5 minutes before drinking his cola?

.....

.....

1 mark

(b) Harry says cola affects his heart rate.

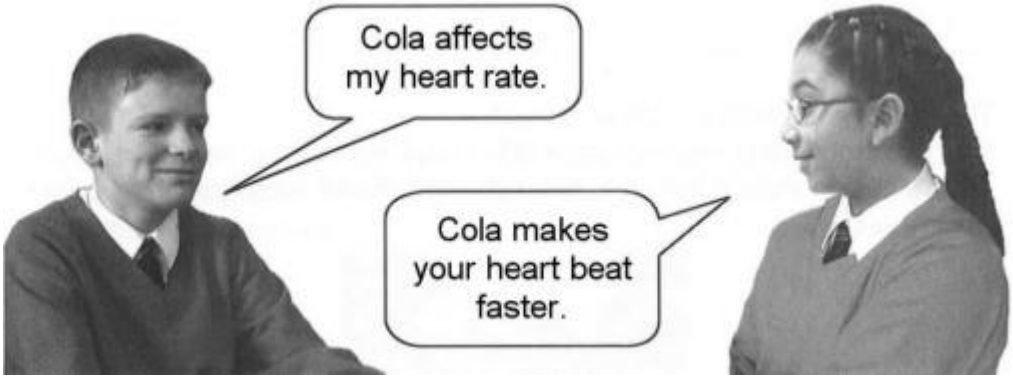
What evidence is there in the graph to support his idea that cola affects his heart rate?

.....

.....

1 mark

(c) Harry and Yasmin came to the following conclusions.



Harry Yasmin

Explain why Yasmin's conclusion is better than Harry's conclusion.

.....

.....

1 mark

(d) Yasmin said, "We should also measure Harry's heart rate after he drinks fizzy water".

How would measuring Harry's heart rate after he drinks fizzy water improve the investigation?

.....

.....

1 mark
maximum 4 marks

Q1.

(a) any **one** from

- to allow his resting pulse rate to be determined
  accept 'to get a reading of heart rate when relaxed'

- to compare his pulse before and after drinking the cola
  accept 'to make sure his heart rate was steady'

- so he would know if it had changed
  accept 'to see if there was any change'
  accept 'so he could find out his normal heart rate'
  accept 'to get a reliable reading of his pulse rate'
  accept 'to find the average'
  'he wanted to know his heart rate before he drank his cola' is insufficient as this is given in the question

1 (L5)

(b) his heart rate went up after he drank some cola
*accept 'his heart rate **or** pulse went up'*
accept 'it shows his heart rate went up'
accept 'the graph shows his heart rate changed'
*accept 'the line goes up **or** changed'*
accept 'the points get higher'

1 (L5)

(c) any **one** from

- Yasmin's conclusion describes how it affected his heart rate whil Harry's just says it affected his heart rate
  *accept 'Yasmin is more specific **or** more detailed'*

- Yasmin's conclusion describes the direction of the change in his heart rate
  accept 'Harry doesn't say what the effect is'

- Yasmin's conclusion describes how his heart rate changed
  accept 'Harry's conclusion could mean that his heart rate was slower'

1 (L6)

(d) any **one** from

- drinking fizzy water would be a control

- it would enable them to see whether fizziness **or** sparkling drinks would have an effect

- it would allow them to see if carbon dioxide **or** carbonation has an effect
  accept 'you would know if it was bubbles that had an effect'

'to see if something else had an effect' is insufficient

- it would show it was something else in cola that had an effect

  accept 'using more
  ***or** different drinks would give more evidence'*

1 (L6)

[4]